

Boundary corrections and finite responses for the localized Weil form

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Abstract

This research note gives an explicit Neumann boundary representation for the gamma part of the localized Weil form. The correction is bounded and noncompact, has essential norm $\pi/2$, and admits controlled tails on finite input spaces. An image expansion retains the endpoint singularities and gives an exponentially decreasing remainder in operator norm at fixed interval length. An unconditional high-frequency lower bound allows exact Schur elimination; standard residual identities then give lower and upper response enclosures. We record the resulting localized positivity criterion and its RH-equivalent formulation over unbounded interval lengths, using the classical Weil criterion. The arithmetic sign of the response remains unresolved. No new positivity interval or practical complexity bound is established.

1 Explicit boundary formulas and their scope

The localized gamma operator differs from its diagonal Neumann comparison by a bounded boundary correction. Its two endpoint singularities prevent approximation by finite mass sums in operator norm, while its action on any fixed finite input space has a controlled tail. This note records the kernel, its essential norm, and two complementary truncation estimates: finite mass sums control the input columns, and grouped images retain the singularities while controlling the remainder in operator norm.

An unconditional positive high sector then permits exact Schur elimination, retaining every coupling to the low modes. Standard residual identities give lower and upper bounds for that response. These calculations are developed from [8]; they provide explicit formulas and error bounds for this operator, without a general priority claim.

Localized Weil positivity, finite-codimension positivity and variational formulations have substantial antecedents; see [1, Section 1]. Connes–Consani–Moscovici give a localized form core, explicit matrices and finite restrictions, whose smallest eigenvalues recover the localized lower bound [2, Proposition 3.4 and Sections 4–5]. Their Sections 7–8 also discuss the prolate approximation to the lowest eigenfunction. The comparison principles, localization and Schur machinery are established tools; the particular boundary and remainder formulas are the focus here. The later matrix conjecture records the remaining arithmetic sign through the classical Weil criterion. Neither that reformulation nor the numerical illustration establishes a new positivity interval or a computational advantage over existing restrictions.

2 The arithmetic operator and an unconditional cutoff

Use $X_L = L^2(0, L)$, zero extension E_L , and $\widehat{F}(\tau) = \int_{\mathbb{R}} F(x) e^{-i\tau x} dx$. Inner products are antilinear in the first slot. Set

$$a_k = 2k + \frac{1}{2}, \quad b(s) = \sum_{k=0}^{\infty} \frac{2}{a_k} \frac{s}{a_k^2 + s} = \operatorname{Re} \psi\left(\frac{1}{4} + i\sqrt{s}/2\right) - \psi\left(\frac{1}{4}\right), \quad (2.1)$$

$$w_0 = \psi\left(\frac{1}{4}\right) - \log \pi, \quad \mathcal{D}_L = \left\{ f \in X_L : \int_{\mathbb{R}} \log(2 + |\tau|) |\widehat{E_L f}(\tau)|^2 d\tau < \infty \right\}. \quad (2.2)$$

Let T_L be the nonnegative operator associated with the closed form

$$t_L[f] = \frac{1}{2\pi} \int_{\mathbb{R}} b(\tau^2) |\widehat{E_L f}(\tau)|^2 d\tau.$$

For $d > 0$, $U_d F(x) = F(x - d)$ and $V_d = E_L^* U_d E_L$. Put $c_L(x) = \cosh((x - L/2)/2)$ and $s_L(x) = \sinh((x - L/2)/2)$. The bounded remainder and the full operator are

$$R_L = w_0 I + 2c_L c_L^* - 2s_L s_L^* - \sum_{\substack{p \text{ prime, } m \geq 1 \\ m \log p < L}} (\log p) p^{-m/2} (V_{m \log p} + V_{m \log p}^*), \quad (2.3)$$

$$W_L = T_L + R_L, \quad Q_L[f] = t_L[f] + \langle f, R_L f \rangle. \quad (2.4)$$

Here $vv^* f = v \langle v, f \rangle$. The sum includes every active prime power. All sums in R_L are finite. In the additive normalization of the Weil functional in [1, Section 1.1], Q_L is exactly the Weil form of $E_L f$. The normalization and its spatial-to-Fourier bridge are detailed in [9, Sections 2–3]. The constant, both poles and all prime powers are essential.

The digamma expansion [3] gives $b(\tau^2) = \log |\tau| - \log 2 - \psi(1/4) + O(\tau^{-2})$. Consequently $\mathcal{D}_L$ is the form domain and $C_c^\infty(0, L)$ is a form core. The support contraction followed by mollification proves the core assertion in this logarithmic Fourier norm [9, Section 3]. Bounded perturbation then defines the self-adjoint, semibounded W_L on $\operatorname{Dom} T_L$.

Let H_N and H_D denote the Neumann and Dirichlet interval Laplacians. Minimizing

$$\frac{2}{a} \left(\|E_L f - u\|_{L^2(\mathbb{R})}^2 + a^{-2} \|u'\|_{L^2(\mathbb{R})}^2 \right)$$

over $u \in H^1(\mathbb{R})$ gives $(2/a)s/(a^2 + s)$ in Fourier space. Restricting u to the interval lowers this minimum; allowing only zero extensions of $H_0^1(0, L)$ competitors raises it. Summation proves the closed-form bracket

$$b(H_N) \leq T_L \leq b(H_D). \quad (2.5)$$

The positive resolvent representation belongs to the complete Bernstein function framework of [4]. For fractional powers of the Laplacian, the corresponding comparisons of the restricted form with the spectral Dirichlet and Neumann forms are established in [5, Theorem 2] and [6, Theorem 3], respectively. Those theorems concern fractional powers; the variational argument above proves (2.5) for this digamma multiplier. The comparison does not assign Neumann boundary conditions to the actual whole-line fields.

Write

$$e_j(x) = \nu_j \cos(j\pi x/L), \quad \nu_0 = L^{-1/2}, \quad \nu_j = \sqrt{2/L} \ (j > 0), \quad b_j = b((j\pi/L)^2).$$

Let $P_h = \sum_{j=0}^{h-1} e_j e_j^*$. For $q \in (0, 1)$ define

$$r_h(q) = \frac{q}{1-q} \left(2 - q^{\lfloor (h-1)/2 \rfloor} - q^{\lceil (h-1)/2 \rceil} \right),$$

and specify

$$\beta_L = -w_0 + 2 \sinh(L/2) - L + \sum_{\log p < L} (\log p) r_{\lceil L/\log p \rceil}(p^{-1/2}). \quad (2.6)$$

The prime bound follows by decomposing translation by $\log p$ into finite chains and bounding each symmetric correlation matrix by its largest row sum r_h : successive row sums differ by $q^i - q^{h-i}$, so the maximum occurs centrally. The odd pole costs at most $2\|s_L\|^2 = 2 \sinh(L/2) - L$; the even pole is nonnegative. Thus

$$W_L \geq b(H_N) - \beta_L I. \quad (2.7)$$

Fix $\delta_0 = 1/16$. For any $L > 0$, call an integer $N \geq 1$ admissible if

$$b_N > \beta_L + \delta_0, \quad P = P_N, \quad Q = I - P. \quad (2.8)$$

Such cutoffs exist because $b_j \rightarrow \infty$. At integer lengths, a terminating rational-enclosure search can select an admissible $N(n)$ without deciding equality at a smaller index [9, Section 4]. This is an optional way to implement the construction. The mathematical family retains N as an explicit parameter.

Let H be the self-adjoint operator associated with the high form Q_L restricted to $Q\mathcal{D}_L$. Then $H \geq \delta_0 I$ unconditionally, so H^{-1} is well defined with norm at most δ_0^{-1} . No invariance of QX_L under W_L is assumed.

3 An explicit boundary operator

The following representation makes that coupling accessible. For $a > 0$ set

$$u_a(x) = e^{-ax}, \quad v_a(x) = e^{-a(L-x)}, \quad r_a = e^{-aL}, \quad K_a = \frac{u_a u_a^* + v_a v_a^* + r_a(u_a v_a^* + v_a u_a^*)}{1 - r_a^2}.$$

Theorem 3.1 (Boundary representation). *The positive finite sums $K^{(J)} = \sum_{k < J} K_{a_k}$ converge strongly to a bounded positive operator K_L satisfying*

$$T_L = b(H_N) + K_L, \quad \text{Dom } T_L = \text{Dom } b(H_N), \quad \mathcal{D}_L = \text{Dom } b(H_N)^{1/2}. \quad (3.1)$$

Finite cosine sums are an operator core. The essential norm is $\|K_L\|_{\text{ess}} = \pi/2$, so finite mass sums do not converge in operator norm. Nevertheless, for the first h cosine modes,

$$\|(K_L - K^{(J)})P_h\| \leq \eta_{h,J} := \sqrt{\frac{2(2h-1)}{L}} (a_J^{-1/2} + a_J^{-3/2}) \rightarrow 0. \quad (3.2)$$

Proof. The compressed free Green function is $e^{-a|x-y|}/(2a)$; the Neumann Green function is

$$\frac{\cosh(a \min(x, y)) \cosh(a(L - \max(x, y)))}{a \sinh(aL)}.$$

Their difference multiplied by $2a$ is K_a . Subtracting the corresponding resolvent energies proves the finite-tower version of (3.1). The coefficient matrix of K_a is positive.

Let $n(t) = e^{-t/2}/(1 - e^{-2t})$. Summing the kernels gives $n(x+y) + n(2L-x-y)$ plus a bounded kernel. Since $n(t) = (2t)^{-1} + O_L(1)$ on $(0, 2L]$, this is one half of each endpoint Carleman kernel, modulo a bounded-kernel, hence compact, operator. The weighted Schur test bounds the Carleman operator by π . The partial towers have the same uniform kernel bound, so monotone convergence yields a bounded strong limit and the asserted domains.

Splitting the interval in half makes the two endpoint operators a direct sum modulo compact operators. Each has essential norm π before the factor $1/2$: normalized functions proportional to $t^{-1/2}$ on $[1, R]$, then dilated toward zero, have limiting Rayleigh quotient π and tend weakly to zero. This also proves the lower essential-norm bound; see the Carleman background in [7].

Direct integration, with $\omega_j = j\pi/L$, gives

$$K_a e_j = \frac{\nu_j a}{a^2 + \omega_j^2} (u_a + (-1)^j v_a). \quad (3.3)$$

Its norm is at most $\sqrt{2}\nu_j a^{-3/2}$. Sum the decreasing series using $\sum_{k \geq j} a_k^{-3/2} \leq a_j^{-3/2} + a_j^{-1/2}$, then sum squared column bounds over $j < h$ to obtain (3.2). $\square$

In particular all low basis vectors lie in $\text{Dom } W_L$. The boundary entries vanish for opposite parities; for equal parities they are

$$\langle e_i, K_L e_j \rangle = 2\nu_i \nu_j \sum_{k \geq 0} \frac{a_k^2 (1 - (-1)^j e^{-a_k L})}{(a_k^2 + \omega_i^2)(a_k^2 + \omega_j^2)}. \quad (3.4)$$

The remaining entries are elementary integrals of the rank-two pole kernel and of translated cosines on their overlap intervals. These formulas specify the operator using gamma and prime data, independently of RH. For a fixed finite compression the entry series has a sharper $O(J^{-1})$ mass tail; for residual norms, errors from a mass approximation are controlled by the full-column tail (3.2) [9, Section 6]. An alternative image expansion sums the gamma masses exactly before truncation. After retaining the singular endpoint images, its remaining kernel has an explicit exponentially decreasing operator-norm bound [9, Section 5]. Evaluating the resulting columns and residual Grams still requires control of the endpoint singularities and integration errors.

4 Exact elimination and localized positivity

For each L and admissible N , use $P = P_N$, $Q = I - P$ and define

$$A = PW_L P, \quad B = QW_L P, \quad H = QW_L Q, \quad \text{Dom } H = Q \text{Dom } b(H_N).$$

Here A is an N by N Hermitian matrix, $B : PX_L \rightarrow QX_L$ is bounded, and H is the operator associated with the high form above; the displayed compression and domain follow from (3.1). Define the finite matrix

$$S_{L,N} = A - B^* H^{-1} B, \quad S_{n,N} := S_{L,N}|_{L=n}. \quad (4.1)$$

For a fixed admissible pair (L, N) , abbreviate $S_L = S_{L,N}$. Its entries involve an infinite-dimensional inverse. It is a finite matrix with explicitly specified, approximable entries, not a formula involving only finitely many elementary operations.

The operators defining $S_{n,N}$ use no zeta zeros or assumption on their location. Their remaining sign is expressed by the following conjecture.

Conjecture 4.1 (Finite response). For every positive integer n and every admissible integer N , the matrix $S_{n,N}$ defined in (4.1) is positive semidefinite.

Equivalently, for each n it suffices to use one admissible N , as the next theorem shows. No enclosure routine is part of the conjecture. In energy language, allowing the positive high modes to respond to a low-mode input never removes more than the low input's original energy.

Theorem 4.2 (Localized positivity and the Weil criterion). *For each $L > 0$ and each admissible N , $Q_L \geq 0$ on $\mathcal{D}_L$ if and only if $S_{L,N} \succeq 0$. Consequently Conjecture 4.1 is equivalent to RH.*

Proof. Write $f = p + h$, $p = Pf$, $h = Qf$. Completing the high square gives the identity of closed forms

$$Q_L[p + h] = \|H^{1/2}(h + H^{-1}Bp)\|^2 + \langle p, S_L p \rangle. \quad (4.2)$$

Every term is defined: $p \in \text{Dom } W_L$, $h \in \text{Dom } H^{1/2}$, and $H^{-1}Bp \in \text{Dom } H$. If $S_L \succeq 0$, this proves positivity. Conversely, choose $h = -H^{-1}Bp$ to recover exactly $\langle p, S_L p \rangle$.

RH implies Weil positivity on smooth compactly supported functions. The form core extends it to $\mathcal{D}_L$, so every $S_{n,N} \succeq 0$. For the converse, every compact support can be translated into $(0, n)$ for some integer n . The Weil form is translation invariant: its gamma multiplier and prime correlations are invariant, and the pole kernel is $2 \cosh((x - y)/2)$. Zero extension to a larger interval preserves all these terms. Thus positivity of $S_{n,N}$ for one admissible N at each n implies Weil positivity on all required test functions, and hence RH by the classical criterion [1]. $\square$

No uniform gap in n , common source on all intervals, or compatible limit of the matrices is required. The dimensions can grow without bound.

Reflection $\mathcal{J}f(x) = f(L - x)$ preserves the form and the cosine cutoff: $\mathcal{J}e_j = (-1)^j e_j$ and $\mathcal{J}V_d \mathcal{J} = V_d^*$. Thus H^{-1} and the response respect parity, and $S_L = S_L^{\text{even}} \oplus S_L^{\text{odd}}$. Both blocks must be nonnegative; their dimensions are $\lceil N/2 \rceil$ and $\lfloor N/2 \rfloor$. Trials respecting parity retain this split in the enclosures below [9, Sections 6–7].

5 Lower and upper response enclosures

Fix L and an admissible N throughout the construction below. Put $d_j = b_j - \beta_L$ for $j \geq N$, and let $\Lambda e_j = d_j e_j$ on QX_L . Then $H \geq \Lambda \geq \delta_0 I$. For any finite-input trial map $Y : PX_L \rightarrow \text{Dom } H$, set $C(Y) = (P - Y)^* W_L (P - Y)$ and $R_Y = B - HY$. Expansion gives the variational identity

$$S_L = C(Y) - R_Y^* H^{-1} R_Y, \quad 0 \preceq R_Y^* H^{-1} R_Y \preceq R_Y^* \Lambda^{-1} R_Y. \quad (5.1)$$

It explains why the energy of a trial is an upper bound until its remaining response is subtracted. For $K > N$, let $P_{N,K}$ project onto $e_N, \dots, e_{K-1}$ and define the Galerkin trial by the positive finite system

$$(P_{N,K} H P_{N,K}) Y_K = P_{N,K} B, \quad \text{ran } Y_K \subset P_{N,K} X_L.$$

Its residual $R_K = B - H Y_K$ is supported in cosine modes $j \geq K$.

Theorem 5.1 (Convergent enclosures). *The finite Hermitian matrices*

$$\begin{aligned} L_{L,N,K} &= A - B^* Y_K - d_K^{-1} R_K^* R_K, \\ U_{L,N,K} &= A - B^* Y_K \end{aligned} \quad (5.2)$$

satisfy $L_{L,N,K} \preceq S_L \preceq U_{L,N,K}$ and converge to S_L in norm as $K \rightarrow \infty$. Their gap is exactly $d_K^{-1} R_K^ R_K$.*

Proof. Galerkin orthogonality gives $Y_K^* H Y_K = B^* Y_K = Y_K^* B$ and $P_{N,K} R_K = 0$. Apply (5.1) and $\Lambda^{-1} \leq d_K^{-1} I$ on the remaining modes to obtain the bounds. Write $H = \Lambda + E$, where $E = Q(K_L + R_L + \beta_L I)Q$ is bounded. Coercivity gives $\|Y_K\| \leq \delta_0^{-1} \|B\|$, while $R_K = (I - P_K)(B - E Y_K)$. Consequently

$$\|U_{L,N,K} - L_{L,N,K}\| \leq \frac{\|B\|^2(1 + \|E\|/\delta_0)^2}{d_K} \rightarrow 0.$$

Each enclosure's distance from S_L is at most this gap. Since $d_K = \log K + O_L(1)$, this is a crude $O_L(1/\log K)$ bound. $\square$

The column tail (3.2) controls finite mass approximations to B and $H Y_K$. The residual Gram must include their interference with the pole and prime columns. Positive eigenvalues of an upper enclosure alone are not a positivity certificate.

Corollary 5.2 (An equivalent tolerance formulation). *Conjecture 4.1, and hence RH , is equivalent to*

$$\begin{aligned} \forall n, m \in \mathbb{N}, \quad \forall N \text{ admissible at } n, \quad \exists K > N : \\ L_{n,N,K} + 2^{-m} I \succ 0, \end{aligned} \tag{5.3}$$

where $L_{n,N,K}$ is (5.2) with $L = n$.

Proof. If $S_{n,N} \succeq 0$, choose K with $\|L_{n,N,K} - S_{n,N}\| < 2^{-m}$, using Theorem 5.1. Conversely $S_{n,N} \succeq L_{n,N,K} \succ -2^{-m} I$ for every m implies $S_{n,N} \succeq 0$. This avoids assuming that a semidefinite limit is detected by an unshifted, finite strict certificate. $\square$

The strict margin makes each true instance amenable to sufficiently accurate enclosures. Arbitrary finite trials and explicit column-error allowances are treated in [9, Section 10]; its Section 9 proves the stronger residual convergence needed there. An optional equivalent comparison of two positive matrices is given in its Section 8. For the same reason as in Theorem 4.2, one may instead choose one admissible N at each n . A finite collection of successful intervals or tolerances does not by itself establish (5.3).

6 Arithmetic sign and limits of the construction

The reduction works for many semibounded operators. For example $W = \begin{pmatrix} 1/2 & 1 \\ 1 & 1 \end{pmatrix}$ has a positive high block $H = 1$ and $W + I > 0$, but its low Schur complement is $-1/2$. A positive shifted source and a positive high sector alone therefore do not imply Conjecture 4.1.

There is a more relevant negative control. Define the Chebyshev function $\psi_{\text{Ch}}(x) = \sum_{p^m \leq x} \log p$ and $E(r) = \psi_{\text{Ch}}(e^r) - e^r + 1$. For $h_f(r) = \text{Re}\langle E_L f, U_r E_L f \rangle$, define the gamma form with its lowest mass removed by

$$t_{\geq 1,L}[f] = \frac{1}{2\pi} \int_{\mathbb{R}} \sum_{k \geq 1} \frac{2}{a_k} \frac{\tau^2}{a_k^2 + \tau^2} |\widehat{E_L f}(\tau)|^2 d\tau.$$

Cancellation of the smooth prime density against the poles and the lowest gamma mass gives

$$Q_L[f] = t_{\geq 1,L}[f] + (4 + w_0)\|f\|^2 - 2 \int_0^L e^{-r/2} h_f(r) dE(r). \tag{6.1}$$

The Stieltjes form holds on $\mathcal{D}_L$; derivatives of h_f are not required. If the prime measure is replaced by $e^r dr$, the last term vanishes. On the unit input $f(x) = \sin(\pi x/2)$ on $(0, 2)$,

$$Q_2^{\text{dens}}[f] < -1 + \frac{26}{125} \frac{(22/7)^2}{4} = -\frac{2979}{6125} < 0.$$

Here $w_0 < -5$ and $\sum_{k \geq 1} 2/a_k^3 \leq 26/125$; smooth approximation preserves negativity. The actual prime discrepancy carries information that the leading density and the positive response architecture cannot replace.

At $L = 1$, an explicit rational certificate permits the alternative cutoff $N = 4$ with $\delta_0 = 1/16$ [9, Section 13]. This certifies only the high sector. Numerical finite-response observations in that section illustrate small eigenvalues but supply no sign certificate. The smallest admissible cutoff $N_{\min}(L)$ already grows doubly exponentially: $\log \log N_{\min}(L) = L/2 + O(\log L)$ as $L \rightarrow \infty$ [9, Section 4]. This concerns the present conservative bound, not an intrinsic minimum dimension for testing the Weil form. Moreover the response estimates supply no positive lower bound on the smallest eigenvalue and hence no uniform working precision for a sign test. No new positivity interval or practical complexity estimate is established.

The boundary and response estimates isolate the arithmetic ordering that remains to be proved. Positive auxiliary constructions alone do not settle it; the exact prime contribution remains essential.

Preparation and verification status

This note and its source investigation were developed with substantial assistance from OpenAI’s ChatGPT and Codex in mathematical analysis, derivation checking, code and exposition. Anthropic’s Claude supplied an additional mathematical review, numerical diagnostics and novelty assessments. Human mathematical verification remains incomplete. The companion [9] expands the arguments and records the checks needed for the unconditional results and the equivalence. Responsibility for the final claims rests with the human author. The repository citation identifies the source revision; a version-specific archival DOI has not yet been supplied.

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